COMPUTER ENGINEERING

QUEZON CITY

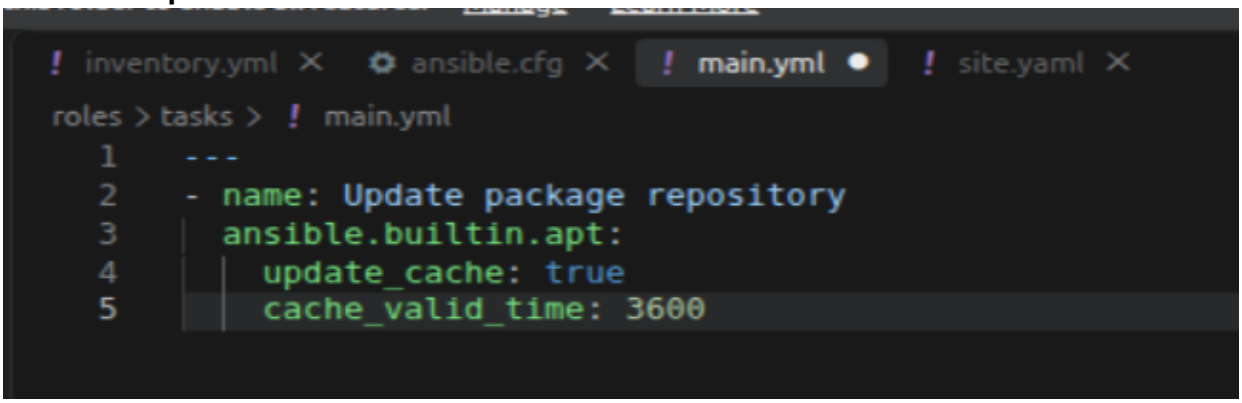
COURSE CODE/TITLE:	CPE212	**SECTION:**	CPE31S2
DATE PERFORMED:	09/24/26	**DATE SUBMITTED:**	09/24/26
NAME:	Ivan Wayne San Juan	**INSTRUCTOR:**	Engr. Jimlord Quejado

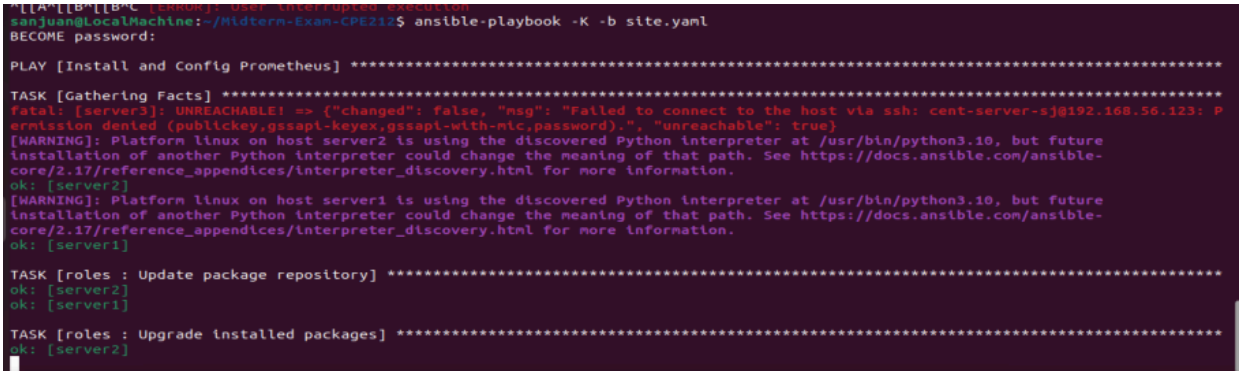
ACTIVITY NO **Midterm Skills Exam**

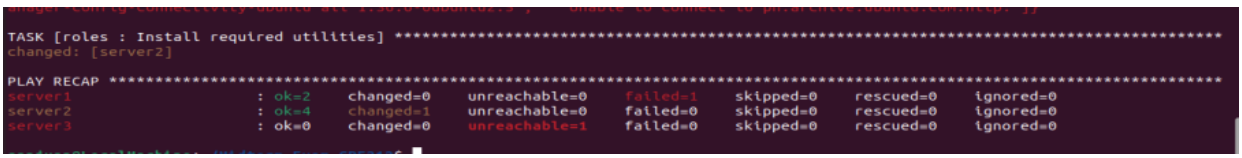
1. PROCEDURE OUTPUT

Sample output (students are allowed to add more steps as needed.)

Task 1: Prepare the Server







Task 2: Create the Prometheus User

inventory.yml | ansible.cfg | main.yml | site.yaml

roles > tasks > main.yml

```yaml
---
# Task 2 - Create the Prometheus User
- name: Create Prometheus system user
  ansible.builtin.user:
    name: "{{ prometheus_user }}"
    system: true
    shell: /usr/sbin/nologin
    create_home: false
  become: true

- name: Verify Prometheus user
  ansible.builtin.command:
    cmd: "id {{ prometheus_user }}"
  register: prometheus_user_check
  changed_when: false

- name: Display Prometheus user information
  ansible.builtin.debug:
    var: prometheus_user_check.stdout
```

```
sanjuan@LocalMachine:~/Midterm-Exam-CPE212$ ansible-playbook -K -b site.yaml
BECOME password:

PLAY [Install and Config Prometheus] ***********************************************************************************************

TASK [Gathering Facts] *************************************************************************************************************
fatal: [server3]: UNREACHABLE! => {"changed": false, "msg": "Failed to connect to the host via ssh: cent-server-sj@192.168.56.123: P
ermission denied (publickey,gssapi-keyex,gssapi-with-mic,password).", "unreachable": true}
[WARNING]: Platform linux on host server2 is using the discovered Python interpreter at /usr/bin/python3.10, but future
installation of another Python interpreter could change the meaning of that path. See https://docs.ansible.com/ansible-
core/2.17/reference_appendices/interpreter_discovery.html for more information.
ok: [server2]
[WARNING]: Platform linux on host server1 is using the discovered Python interpreter at /usr/bin/python3.10, but future
installation of another Python interpreter could change the meaning of that path. See https://docs.ansible.com/ansible-
core/2.17/reference_appendices/interpreter_discovery.html for more information.
ok: [server1]

TASK [roles : Create Prometheus system user] ***************************************************************************************
fatal: [server1]: FAILED! => {"msg": "The task includes an option with an undefined variable.. 'prometheus_user' is undefined\n\nThe
 error appears to be in '/home/sanjuan/Midterm-Exam-CPE212/roles/tasks/main.yml': line 3, column 3, but may\nbe elsewhere in the fil
e depending on the exact syntax problem.\n\nThe offending line appears to be:\n\n# Task 2 - Create the Prometheus User\n- name: Crea
te Prometheus system user\n  ^ here\n"}
fatal: [server2]: FAILED! => {"msg": "The task includes an option with an undefined variable.. 'prometheus_user' is undefined\n\nThe
 error appears to be in '/home/sanjuan/Midterm-Exam-CPE212/roles/tasks/main.yml': line 3, column 3, but may\nbe elsewhere in the fil
e depending on the exact syntax problem.\n\nThe offending line appears to be:\n\n# Task 2 - Create the Prometheus User\n- name: Crea
te Prometheus system user\n  ^ here\n"}

PLAY RECAP *************************************************************************************************************************
server1                    : ok=1    changed=0    unreachable=0    failed=1    skipped=0    rescued=0    ignored=0
server2                    : ok=1    changed=0    unreachable=0    failed=1    skipped=0    rescued=0    ignored=0
server3                    : ok=0    changed=0    unreachable=1    failed=0    skipped=0    rescued=0    ignored=0
```

Task 3: Create Prometheus Directories

roles > tasks > main.yml

```yaml
---
# Task 3 - Create Prometheus Directories
- name: Create Prometheus configuration directory
  ansible.builtin.file:
    path: "{{ prometheus_config_dir }}"
    state: directory
    owner: "{{ prometheus_user }}"
    group: "{{ prometheus_group }}"
    mode: "0755"
  become: true

- name: Create Prometheus data directory
  ansible.builtin.file:
    path: "{{ prometheus_data_dir }}"
    state: directory
    owner: "{{ prometheus_user }}"
    group: "{{ prometheus_group }}"
    mode: "0755"
  become: true

- name: Verify Prometheus directories
  ansible.builtin.command:
    cmd: "ls -ld {{ prometheus_config_dir }} {{ prometheus_data_dir }}"
  register: prometheus_directories
  changed_when: false

- name: Display Prometheus directory information
  ansible.builtin.debug:
    var: prometheus_directories.stdout
```

```
sanjuan@LocalMachine:~/Midterm-Exam-CPE212$ ansible-playbook -K -b site.yaml
BECOME password:

PLAY [Install and Config Prometheus] ***********************************************************************************************

TASK [Gathering Facts] *************************************************************************************************************
fatal: [server3]: UNREACHABLE! => {"changed": false, "msg": "Failed to connect to the host via ssh: cent-server-sj@192.168.56.123: P
ermission denied (publickey,gssapi-keyex,gssapi-with-mic,password).", "unreachable": true}
[WARNING]: Platform linux on host server2 is using the discovered Python interpreter at /usr/bin/python3.10, but future
installation of another Python interpreter could change the meaning of that path. See https://docs.ansible.com/ansible-
core/2.17/reference_appendices/interpreter_discovery.html for more information.
ok: [server2]
[WARNING]: Platform linux on host server1 is using the discovered Python interpreter at /usr/bin/python3.10, but future
installation of another Python interpreter could change the meaning of that path. See https://docs.ansible.com/ansible-
core/2.17/reference_appendices/interpreter_discovery.html for more information.
ok: [server1]

TASK [roles : Create Prometheus configuration directory] ***************************************************************************
fatal: [server1]: FAILED! => {"msg": "The task includes an option with an undefined variable.. 'prometheus_config_dir' is undefined\
n\nThe error appears to be in '/home/sanjuan/Midterm-Exam-CPE212/roles/tasks/main.yml': line 3, column 3, but may\nbe elsewhere in t
he file depending on the exact syntax problem.\n\nThe offending line appears to be:\n\n# Task 3 - Create Prometheus Directories\n- n
ame: Create Prometheus configuration directory\n  ^ here\n"}
fatal: [server2]: FAILED! => {"msg": "The task includes an option with an undefined variable.. 'prometheus_config_dir' is undefined\
n\nThe error appears to be in '/home/sanjuan/Midterm-Exam-CPE212/roles/tasks/main.yml': line 3, column 3, but may\nbe elsewhere in t
he file depending on the exact syntax problem.\n\nThe offending line appears to be:\n\n# Task 3 - Create Prometheus Directories\n- n
ame: Create Prometheus configuration directory\n  ^ here\n"}

PLAY RECAP *************************************************************************************************************************
server1                    : ok=1    changed=0    unreachable=0    failed=1    skipped=0    rescued=0    ignored=0
server2                    : ok=1    changed=0    unreachable=0    failed=1    skipped=0    rescued=0    ignored=0
server3                    : ok=0    changed=0    unreachable=1    failed=0    skipped=0    rescued=0    ignored=0
```

Task 4: Download Prometheus

roles > tasks > ! main.yml

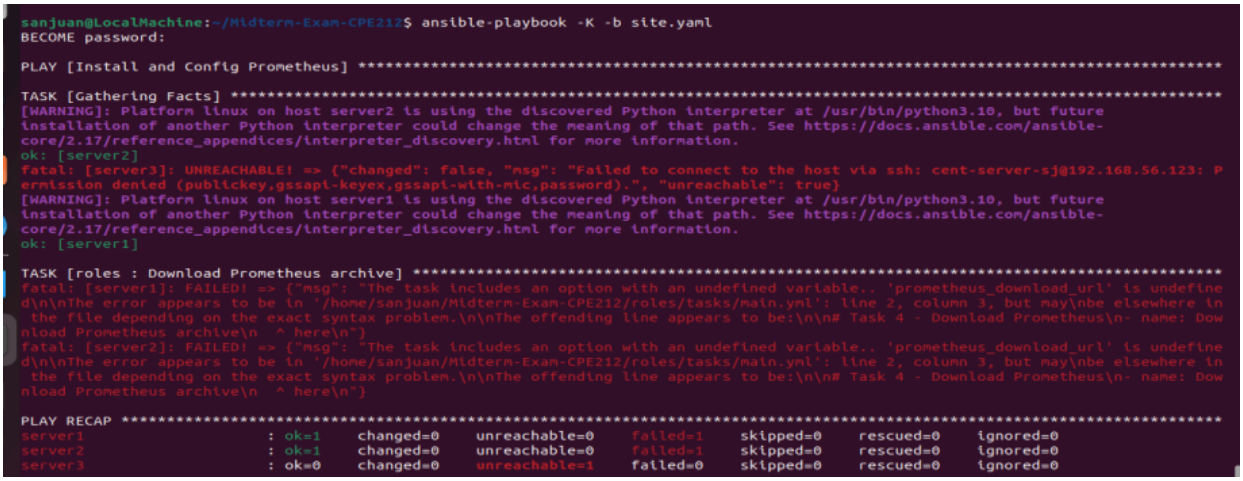
```yaml
# Task 4 - Download Prometheus
- name: Download Prometheus archive
  ansible.builtin.get_url:
    url: "{{ prometheus_download_url }}"
    dest: "{{ prometheus_archive }}"
    mode: "0644"

- name: Verify downloaded Prometheus archive
  ansible.builtin.stat:
    path: "{{ prometheus_archive }}"
  register: prometheus_archive_check

- name: Extract Prometheus archive
  ansible.builtin.unarchive:
    src: "{{ prometheus_archive }}"
    dest: "/tmp"
    remote_src: true
    # NOTE: prometheus_extract_dir must be the FULL extracted path
    # (e.g. /tmp/prometheus-2.53.0.linux-amd64), not just a folder name,
    # or this idempotency check never matches and re-extracts every run.
    creates: "{{ prometheus_extract_dir }}"

- name: Verify extracted Prometheus directory
  ansible.builtin.command:
    cmd: "ls -la {{ prometheus_extract_dir }}"
  register: prometheus_extracted_files
  changed_when: false

- name: Display extracted Prometheus files
  ansible.builtin.debug:
    var: prometheus_extracted_files.stdout
```



CONCLUSION:

Working through this task reinforced the importance of mastering Ansible. Converting manual steps into automated workflows saves significant time and effort.

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Copy and paste the questions from the lab and include your answers after.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: _____

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: _____

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: _____

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

I understand that any false, incomplete, or misleading disclosure may be subject to investigation in accordance with due process and may result in sanctions for violation of existing T.I.P. policies.



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